Intellectual Disability, X-linked Syndromic 35 (MRXS35): A Comprehensive Disease Characterization

Disease: Intellectual Disability, X-linked Syndromic 35 (MRXS35) **MONDO ID:** MONDO:0030908 · **OMIM:** #300998 · **Gene:** *RPL10* (uL16/QM), Xq28 **Category:** Genetic (X-linked recessive ribosomopathy) **Report type:** Aggregated disease-level synthesis from primary literature and reference databases

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Summary

Intellectual Disability, X-linked Syndromic 35 (MRXS35; OMIM #300998, MONDO:0030908) is an ultra-rare X-linked recessive ribosomopathy caused by hemizygous hypomorphic missense variants in *RPL10*, the gene encoding the 60S large-subunit ribosomal protein uL16 (also known as QM) at Xq28. The protein is essential for joining of the 40S and 60S ribosomal subunits during translation initiation, and it sits in close proximity to the peptidyl transferase center of the ribosome. Pathogenic variants reduce functional RPL10, lowering bulk protein synthesis and increasing neuronal apoptosis during brain development — the mechanistic basis for microcephaly and intellectual disability.

Affected males present with a syndromic constellation dominated by intellectual disability/ developmental delay (near-universal), and a variable, partly genotype-dependent combination of autism spectrum disorder, epilepsy, microcephaly (~50% of cases, frequently progressive/ postnatal), cerebellar signs, facial dysmorphism, short stature, hypotonia, and congenital anomalies affecting the eyes (retinal degeneration), ears (hearing loss), heart, and genitourinary and gastrointestinal systems. Reported pathogenic alleles include p.K78E (c.232A>G), p.R32L (c.95G>T), p.R116Q (c.347G>A), and the original autism-associated C-terminal substitutions p.L206M and p.H213Q. Inheritance is typically from unaffected carrier mothers who exhibit skewed (nonrandom) X-inactivation, consistent with X-linked recessive transmission; de novo occurrence is also documented.

There is no gene-specific or disease-modifying therapy. Management is supportive and multidisciplinary — antiepileptic drugs for seizures, developmental/rehabilitative therapies, and surveillance for ophthalmologic, audiologic, cardiac, and genitourinary complications — coupled with molecular diagnosis by exome/genome or gene-panel sequencing and genetic counseling for at-risk families. Notably, the *same* gene harbors a mechanistically distinct recurrent **somatic** mutation, p.R98S, in ~8% of pediatric T-cell acute lymphoblastic leukemia, which activates JAK-STAT signaling — a clean illustration that germline hypomorphic and somatic oncogenic *RPL10* lesions produce entirely different diseases.

Key Findings

Finding 1 — MRXS35 is caused by hemizygous missense variants in *RPL10* (uL16), an X-linked ribosomopathy

MRXS35 (OMIM #300998) is caused by hemizygous missense variants in **RPL10** (HGNC:10298; Entrez Gene 6134; Ensembl ENSG00000147403; gene-MIM 312173; cytoband Xq28; aliases QM, uL16, AUTSX5, L10, DXS648, NOV). The gene was first implicated in neurodevelopmental disease through autism families, where two C-terminal missense substitutions (p.L206M and p.H213Q) were identified: *"We have identified two missense mutations in the ribosomal protein gene RPL10 located in Xq28 in two independent families with autism"* (PMID: 16940977). The syndromic microcephaly/ID phenotype was subsequently defined by the identification of *"a novel missense mutation in the gene encoding 60S ribosomal protein L10 (RPL10)"* — the p.K78E allele (PMID: 25316788).

The recurrent pathogenic missense alleles map to functionally critical regions of the protein. The N-terminal variants (p.R32L, p.K78E) fall within the 28S/25S rRNA-binding region near the peptidyl transferase center, while p.L206M and p.H213Q lie in the C-terminal region. Inheritance follows an X-linked recessive paradigm: *"the p.K78E change segregated with disease under an X-linked recessive paradigm while, consistent with causality, carrier females exhibited skewed X inactivation"* (PMID: 25316788). The normal molecular role of the protein — it is *"required for the joining of the 40S and 60S subunits"* (PMID: 9443083) — anchors the ribosomopathy mechanism. Zebrafish complementation established that p.K78E is a loss-of-function allele.

Ontology anchors: Gene HGNC:10298 (*RPL10*); disease MONDO:0030908; GO:0022618 (ribonucleoprotein complex assembly), GO:0042254 (ribosome biogenesis), GO:0006412 (translation).

Finding 2 — *RPL10* is highly intolerant to loss-of-function and missense variation

Population constraint metrics from gnomAD confirm that *RPL10* is under strong purifying selection, consistent with an essential housekeeping gene. gnomAD v2 constraint values for ENSG00000147403 are: **pLI** = **0.997** (loss-of-function intolerant), **observed/expected LoF** = **0.064** (1 observed vs. 15.7 expected; LOEUF/oe_lof_upper = 0.30), **LoF-Z** = **3.15**, and **missense-Z** = **4.08** (oe_mis = 0.29), while the synonymous-Z of 0.37 confirms the model is well-calibrated (neutral for silent variation). The gene spans chrX:154,389,955–154,409,168 (GRCh38), Xq28.

The practical consequence is that pathogenic *RPL10* missense variants are essentially absent from population reference databases: for one reported pathogenic allele, the *"variant, not reported in gnomAD"* (PMID: 35876338). This strong constraint supports ACMG criterion PM2 (absent/rare in population databases) for candidate variants and explains why only hypomorphic (partial loss-of-function) missense alleles — rather than complete null alleles — are compatible with viability and thus observed in patients.

Constraint metric (gnomAD v2)	Value	Interpretation
pLI	0.997	LoF-intolerant
oe LoF (LOEUF)	0.064 (0.30)	Strong LoF depletion
LoF-Z	3.15	Significant constraint
Missense-Z	4.08	Strong missense constraint
Synonymous-Z	0.37	Neutral (calibrated)

Finding 3 — Clinical spectrum: syndromic ID with epilepsy, ASD, microcephaly, dysmorphism, and congenital anomalies

Across ~15–19 reported affected males, the core phenotype comprises **intellectual disability/ developmental delay** (near-universal; 4/4 in the Thevenon 2015 family), **autism spectrum disorder**, **epilepsy** (often early-onset, refractory seizures), and **facial dysmorphism**. MRXS35 is defined as *"an X-linked syndrome presenting with intellectual disability (ID), autism spectrum disorder, epilepsy, dysmorphic features, and multiple congenital anomalies"* (PMID: 35876338).

Microcephaly occurs in approximately half of cases and is often progressive/postnatal: *"microcephaly was observed in approximately half of the cases"* (PMID: 35876338). A family segregating an *RPL10* variant showed a characteristic combination of *"syndromic features including amniotic fluid excess (3/4), microcephaly (2/4), urogenital anomalies (3/4), cerebellar syndrome (2/4), and facial dysmorphism"* (PMID: 25846674), documenting per-feature frequencies. The severe end of the spectrum, associated with p.K78E, includes *"severe microcephaly, seizures, hearing loss, growth retardation, cardiac defects, and dysmorphic facial features"* (PMID: 29066376).

Certain features are variant-associated: retinal degeneration/retinitis pigmentosa is notable with p.R32L, while severe multisystem involvement (hearing loss, cardiac defects) clusters with p.K78E. Onset spans the prenatal/neonatal period (polyhydramnios) to early childhood, and both microcephaly and retinopathy can be **progressive**.

Phenotype	Approx. frequency	Suggested HPO term
Intellectual disability / developmental delay	Near-universal	HP:0001249 / HP:0001263
Facial dysmorphism	Very frequent	HP:0001999
Microcephaly (often progressive/postnatal)	~50%	HP:0000252 / HP:0005484
Epilepsy / seizures	Frequent	HP:0001250
Autism spectrum disorder	Frequent	HP:0000717
Cerebellar signs / ataxia	~50% (2/4)	HP:0001251
Urogenital/genital anomalies (incl. cryptorchidism)	Frequent (3/4)	HP:0000078 / HP:0000028
Polyhydramnios (antenatal)	Frequent (3/4)	HP:0001561
Short stature / growth retardation	Frequent	HP:0004322
Hypotonia	Reported	HP:0001252
Retinal degeneration (variant-associated, p.R32L)	Subset	HP:0000546
Hearing loss (variant-associated, p.K78E)	Subset	HP:0000365
Cardiac defects (variant-associated, p.K78E)	Subset	HP:0001627

Finding 4 — Mechanism: germline hypomorphic variants impair translation and increase apoptosis; distinct somatic R98S drives leukemia

The germline (MRXS35) mechanism is a partial loss of ribosomal function. In a zebrafish model, *rpl10* suppression *"decreases head size in developing morphant embryos, concomitant with reduced bulk translation and increased apoptosis in the brain"* (PMID: 25316788); this phenotype is rescued by wild-type but not mutant human RPL10, establishing p.K78E as loss-of-function. Because uL16 *"is required for the joining of the 40S and 60S subunits"* (PMID: 9443083) and is added late during cytoplasmic 60S maturation, reduced functional protein throttles the supply of translation-competent ribosomes precisely when developing neurons demand high protein synthesis — leading to reduced neuronal output, apoptosis, and microcephaly.

Critically, the same gene harbors a **mechanistically distinct somatic** lesion in cancer. The recurrent p.R98S mutation occurs in ~8% of pediatric T-cell acute lymphoblastic leukemia (T-ALL): *"the recurrent RPL10-R98S mutation in T-cell acute lymphoblastic leukemia (T-ALL)"* (PMID: 30482776). Rather than simple loss of function, R98S produces a gain of oncogenic activity — *"we describe modulation of the JAK-STAT cascade as a novel cancer-promoting activity of a ribosomal mutation"* (PMID: 28744013) — via reduced JAK1 degradation and altered programmed ribosomal frameshifting. R98S also induces oxidative stress (an early proliferation defect), upregulates serine biosynthesis (PSPH) and glycine (PMID: 31186416), and promotes secondary oncogenic mutagenesis including NOTCH1-activating lesions (PMID: 30482776). This germline-vs-somatic contrast underscores that *RPL10* genotype–phenotype relationships are allele-specific and mechanism-specific.

Finding 5 — RPL10/uL16 protein: 214-aa cytoplasmic large-subunit ribosomal protein with resolved structures

The protein product is UniProt **P27635** (human RPL10/uL16): 214 amino acids, cytoplasmic. UniProt annotation states it is a *"Component of the large ribosomal subunit... Plays a role in the formation of actively translating ribosomes... May play a role in the embryonic brain development."* At least 37 experimental PDB entries resolve uL16 within the human 80S ribosome (e.g., 5AJ0, 6OLE, 6OLF, 6OLG, 6OLI, 6OLZ), providing structural context for pathogenic residues. The MRXS35 residues Arg32, Lys78, Leu206, and His213 localize to rRNA-binding/N-terminal and C-terminal regions of the protein, and the protein sits in *"close proximity to the peptidyl transferase active site of the 60S ribosomal subunit"* (PMID: 25316788). This structural proximity to the catalytic center provides a plausible structural rationale for how single missense substitutions perturb translation.

Subcellular localization (GO Cellular Component): GO:0022625 (cytosolic large ribosomal subunit), GO:0005737 (cytoplasm), GO:0005840 (ribosome).

Finding 6 — ClinVar landscape and cross-species orthologs

ClinVar (RefSeq NM_006013.5) holds ~404 *RPL10* records. At the single-nucleotide level, most missense variants are classified as **variants of uncertain significance (VUS)** (e.g., p.Leu36Gln, p.Asp44Tyr, p.Arg32Cys, p.Ile167Val), with a smaller set of Pathogenic/Likely-pathogenic SNVs corresponding to the known MRXS35 alleles. Importantly, many ClinVar entries labeled

"Pathogenic" as copy-number gains/losses are large **multigenic Xq deletions/duplications** that span *RPL10* rather than *RPL10*-specific point mutations — a distinction that matters for interpretation. The disease-relevant residues are *"affecting an evolutionary conserved residue"* (PMID: 35876338), consistent with strong cross-species conservation.

Validated orthologs supporting model-organism work: mouse *Rpl10* (NCBI Gene 110954; MGI:105943; ENSMUSG00000008682); zebrafish *rpl10* (NCBI Gene 336712); *S. cerevisiae* RPL10. Human *RPL10* = NCBI Gene 6134, UniProt P27635 (214 aa).

Finding 7 — Novel p.Arg116Gln allele expands genotype, phenotype, and functional evidence

A 2025 report identified a novel recurrent hemizygous missense variant, **NM_006013.5:c.347G>A, p.Arg116Gln**, in two unrelated Chinese male patients, each maternally inherited: *"the same hemizygous missense RPL10 gene variant (NM_006013.5:c.347G>A, p.Arg116Gln) in each patient, inherited from their respective mothers"* (PMID: 40861044). In vitro functional analysis provided independent loss-of-function evidence: the variant *"reduced the mRNA expression of the RPL10 gene, thereby decreasing synthesis of the RPL10 protein"* (PMID: 40861044).

This report also expanded the neonatal/early phenotype to include *"congenital laryngeal stridor, feeding difficulties, neonatal pneumonia, neonatal hypoglycemia, dysmorphic features, and bilateral cryptorchidism"* (PMID: 40861044), along with short stature, hypotonia, gastrointestinal problems, and craniofacial anomalies. The summarized full MRXS35 spectrum from this paper — ID, psychomotor/speech delay, short stature, craniofacial anomalies, hypotonia, seizures, GI problems, genitourinary anomalies, cardiac anomalies, eye defects, and hearing loss — reinforces the multisystem, prenatal-to-childhood nature of the disorder.

Section-by-Section Report

1. Disease Information

MRXS35 is an ultra-rare, X-linked recessive syndromic intellectual disability disorder — a **ribosomopathy** — caused by variants in *RPL10*. Key identifiers: **OMIM #300998**; **MONDO:0030908**; gene-MIM **312173** (*RPL10*). ICD-11 would fall under 6A00 (Disorders of intellectual development) with a genetic modifier; a specific Orphanet number is not firmly assigned in the retrieved data (the disorder overlaps the "RPL10-related disorder" concept). Common synonyms/alternative names: **MRXS35**; **RPL10-related disorder**; **X-linked intellectual disability, syndromic, 35**; historically linked to **AUTSX5** (autism, X-linked 5) via the same gene. The information in this report is derived from **aggregated disease-level resources** (OMIM, ClinVar, gnomAD, UniProt, PDB) and **individual patient case reports/series** in primary literature, not from EHR-scale patient data.

2. Etiology

Causal factor: monogenic — hemizygous hypomorphic missense variants in *RPL10* (Xq28). **Genetic risk factors:** the causal variants themselves (p.K78E, p.R32L, p.R116Q, p.L206M, p.H213Q); being male (hemizygous) is the principal determinant of clinical expression. **Environmental risk factors:** none established — this is a Mendelian disorder without a recognized environmental contribution. **Protective factors:** in carrier females, skewed X-inactivation favoring the wild-type allele is effectively protective, explaining why carrier mothers are typically unaffected. No dietary/lifestyle protective factors are known. **Gene–environment interactions:** none documented; disease liability is essentially determined by genotype and X-inactivation status.

3. Phenotypes

See Finding 3 table for phenotype types, frequencies, and HPO terms. Phenotype **characteristics**: onset is prenatal/neonatal (polyhydramnios, feeding difficulties, laryngeal stridor, neonatal hypoglycemia) to early childhood (developmental delay, seizures); severity is **variable** and partly genotype-dependent (p.K78E severe; C-terminal autism alleles milder/behavioral); progression is largely **stable** for ID but **progressive** for postnatal microcephaly and retinopathy. **Quality-of-life impact** is substantial where ID is moderate-to-severe, with lifelong dependency, communication impairment, and comorbid epilepsy and behavioral (ASD) burden; formal EQ-5D/SF-36 data are not available for this ultra-rare disorder.

4. Genetic/Molecular Information

Causal gene: *RPL10* (HGNC:10298; NCBI 6134; gene-MIM 312173; Xq28). **Pathogenic variants (germline):** p.K78E (c.232A>G), p.R32L (c.95G>T), p.R116Q (c.347G>A), p.L206M, p.H213Q — all **missense**, all in the RefSeq NM_006013.5 transcript. **Classification:** the recurrent disease alleles are Pathogenic/Likely-pathogenic; the broader ClinVar landscape is VUS-dominant (Finding 6). **Allele frequency:** absent/ultra-rare in gnomAD (Finding 2). **Origin:** germline; typically maternally inherited from carriers with skewed X-inactivation, with de novo cases reported. **Functional consequence:** loss/reduction of function (hypomorphic), demonstrated in zebrafish (p.K78E) and by reduced mRNA/protein for p.R116Q. **Modifier factors:** X-inactivation pattern is the principal modifier in carrier females. **Chromosomal abnormalities:** large Xq28 copy-number gains/losses spanning *RPL10* exist (e.g., MidXq28-duplication syndrome involving FLNA, RPL10, GDI1; and dosage-dependent Xq28 gains where *GDI1* is the likelier driver — PMID: 31090057, PMID: 20004760), but these multigenic CNVs are distinct from *RPL10* point-mutation MRXS35. **Epigenetics:** no disease-specific methylation signature established.

5. Environmental Information

Not applicable. MRXS35 is a monogenic disorder; no environmental toxins, lifestyle factors, or infectious agents are implicated in causation or triggering.

6. Mechanism / Pathophysiology

Ordered causal chain (germline MRXS35):

1. A hemizygous hypomorphic missense variant in *RPL10* (e.g., p.K78E, p.R116Q) **results in** a structurally/functionally impaired uL16 protein (or reduced protein level, as shown for p.R116Q).
2. Reduced functional uL16 **leads to** impaired late cytoplasmic 60S maturation and defective 40S–60S subunit joining (uL16 is *"required for the joining of the 40S and 60S subunits"*).
3. Defective subunit joining **results in** a reduced pool of actively translating 80S ribosomes and **decreased bulk protein synthesis** (demonstrated in zebrafish morphants).
4. Reduced translation in the developing brain **leads to** increased neuronal **apoptosis** and reduced neuronal output (demonstrated).
5. Neuronal loss/reduced proliferation **results in** reduced brain growth → **microcephaly** (often postnatal/progressive) and disrupted neurodevelopment → **intellectual disability, autism, epilepsy** (mechanism inferred from the translation-apoptosis link; clinical correlation strong).
6. **Branch:** tissue-specific translational vulnerability **leads to** variant-associated congenital anomalies — retinal degeneration (p.R32L), hearing loss/cardiac defects (p.K78E), and genitourinary/GI anomalies (inferred).

Pathways/processes: cytoplasmic translation (GO:0006412), ribosomal large subunit assembly/biogenesis (GO:0000027, GO:0042273), apoptotic process (GO:0006915), embryonic brain development. **Protein dysfunction:** partial loss of function of a ribosomal structural protein near the peptidyl transferase center. **Cell types (CL):** neurons (CL:0000540), neural progenitor/neuroblast (CL:0000031), and photoreceptor/cochlear/cardiac cells in variant-specific branches. This contrasts with the **somatic** T-ALL mechanism (p.R98S → JAK-STAT gain-of-function, serine/glycine metabolic rewiring, oncogenic mutagenesis; Finding 4).

7. Anatomical Structures Affected

Primary organ: brain (UBERON:0000955), especially cerebrum and cerebellum (UBERON:0002037; cerebellar signs). **Body system:** central nervous system (UBERON:0001017). **Secondary/variant-associated:** eye/retina (UBERON:0000970 / UBERON:0000966; retinal degeneration), ear/cochlea (hearing loss), heart (UBERON:0000948; cardiac defects), genitourinary tract (cryptorchidism), gastrointestinal tract, and larynx (congenital stridor). **Tissue/cell level:** nervous tissue; neurons and neural progenitors. **Subcellular:** cytoplasm/cytosolic ribosome (GO:0022625). **Lateralization:** bilateral/symmetric involvement (microcephaly, retinopathy, cryptorchidism when bilateral).

8. Temporal Development

Onset: congenital to early childhood; antenatal signs (polyhydramnios) and neonatal features (feeding difficulty, stridor, hypoglycemia) can precede developmental delay. **Onset pattern:** chronic/insidious for neurodevelopment; microcephaly frequently **postnatal and progressive**. **Progression:** intellectual disability is generally stable (non-degenerative), but head circumference and retinal findings can worsen over time. **Course:** chronic, lifelong. **Critical period:** the fetal/early-postnatal window of maximal neurogenesis and neuronal protein-synthesis demand is the period of greatest vulnerability and the theoretical window for intervention.

9. Inheritance and Population

Inheritance: X-linked recessive; affected males, unaffected carrier mothers with skewed X-inactivation; de novo cases reported. **Penetrance:** high/complete in hemizygous males; carrier females usually unaffected due to favorable X-inactivation (rare manifesting carriers possible). **Expressivity:** variable, partly genotype-dependent. **Epidemiology:** ultra-rare; fewer than ~20 well-characterized males reported worldwide; precise prevalence/incidence are not established. **Sex ratio:** strongly male-predominant. **Founder effects/consanguinity:** none established; the p.R116Q allele recurred in two unrelated Chinese families (recurrent mutation, not proven founder). **Carrier frequency:** not defined (variants absent from gnomAD).

10. Diagnostics

Diagnosis is **molecular**. Recommended approach: **whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)**, or a targeted **intellectual disability/XLID gene panel** including *RPL10*; single-gene testing is appropriate when a familial variant is known. **Chromosomal microarray (CMA)** is indicated to detect Xq28 CNVs in the differential (MidXq28-duplication syndrome). Variant interpretation follows **ACMG/AMP** guidelines: absence from gnomAD (PM2), evolutionary conservation, and functional/segregation evidence support pathogenicity; many *RPL10* missense variants remain **VUS**. Supportive workup: brain MRI (microcephaly, cerebellar/structural findings), EEG (epilepsy), ophthalmologic exam (retinopathy), audiology, echocardiography, and growth/endocrine and metabolic evaluation (neonatal hypoglycemia). No specific biochemical biomarker exists. **Differential diagnosis:** other X-linked syndromic IDs (e.g., MECP2-related, KDM5C/Claes-Jensen, USP9X-related, MSL3/Basilicata-Akhtar) and other ribosomopathies. **Screening:** cascade carrier testing in families once the variant is identified; prenatal/preimplantation testing possible for known familial variants. No newborn-screening program exists.

11. Outcome/Prognosis

Survival: the disorder is chronic and non-malignant; life expectancy depends on severity of epilepsy and congenital anomalies (severe p.K78E cases with cardiac defects carry higher risk). No formal survival statistics are available. **Morbidity:** dominated by lifelong intellectual disability, communication impairment, epilepsy, and behavioral (ASD) burden, with additional disability from sensory (visual, auditory) and motor (hypotonia, cerebellar) involvement. **Complications:** refractory seizures, feeding difficulties/aspiration, growth failure, and organ-specific issues. **Recovery:** no reversal; management is supportive. **Prognostic factors:** specific genotype (p.K78E severe end), presence of epilepsy, degree of microcephaly, and multi-organ involvement.

12. Treatment

There is **no disease-modifying or gene-specific therapy**. Management is **supportive and multidisciplinary** (suggested NCIT: Supportive Care Intervention): - **Antiepileptic pharmacotherapy** for seizures (agent selection by seizure type; NCIT: Anticonvulsant Agent). - **Developmental and rehabilitative therapies** — physical, occupational, and speech therapy; special education. - **Surveillance and management** of ophthalmologic (retinopathy), audiologic (hearing loss, hearing aids), cardiac (congenital defect management), genitourinary (cryptorchidism — orchidopexy), and gastrointestinal/feeding complications. - **Nutritional support** for feeding difficulties and growth failure.

No pharmacogenomic guidance, gene therapy, cell therapy, RNA-based therapy, or targeted/immunotherapy is currently available or in trials for MRXS35 (no NCT identifiers retrieved). Personalized/precision approaches remain theoretical.

13. Prevention

Primary prevention is not possible for a spontaneous/inherited Mendelian variant. **Secondary/tertiary prevention** centers on early molecular diagnosis, early developmental intervention, and surveillance to prevent complications (e.g., seizure control, visual/hearing rehabilitation). **Genetic counseling** is central: for carrier mothers, recurrence risk is 50% for sons (affected) and 50% for daughters (carriers). **Reproductive options** include prenatal diagnosis and preimplantation genetic testing for known familial variants, and cascade testing of at-risk female relatives. No immunization, public-health, or environmental interventions apply.

14. Other Species / Natural Disease

RPL10 is **highly evolutionarily conserved.** Orthologs: mouse *Rpl10* (NCBI 110954; MGI:105943; ENSMUSG00000008682); zebrafish *rpl10* (NCBI 336712); *S. cerevisiae* RPL10. No naturally occurring companion-animal or wildlife disease equivalent to MRXS35 is documented in the retrieved data (no OMIA entry established here). Conservation of uL16's role in subunit joining across yeast, fish, and mammals underpins the validity of cross-species functional modeling. No zoonotic or cross-species transmission is relevant (non-infectious genetic disorder).

15. Model Organisms

The principal disease model is the **zebrafish** (*Danio rerio*): morpholino knockdown of *rpl10* reproduces reduced head size, decreased bulk translation, and increased brain apoptosis; complementation (rescue) assays with wild-type vs. mutant human *RPL10* established loss-of-function for p.K78E (PMID: 25316788). This model recapitulates the core microcephaly phenotype and directly demonstrates the translation–apoptosis mechanism. **Yeast** (*S. cerevisiae*) provided foundational evidence that QM/uL16 assembles onto the 60S subunit in the cytoplasm and is required for subunit joining (PMID: 9443083). A validated **mouse** ortholog exists (MGI:105943) and would be the natural system for a conditional/knock-in model, though a published MRXS35-specific mouse model was not identified. **Limitations:** morphant models capture microcephaly/translation defects but not the full multisystem, behavioral (ASD/epilepsy), or progressive retinal spectrum; humanized knock-in mammalian models are a key gap.

Mechanistic Model / Interpretation

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RPL10 hypomorphic missense variant (e.g., p.K78E, p.R116Q, p.R32L)
                       │
                       ▼
Impaired / reduced uL16 protein (loss-of-function; near PTC)
                       │
                       ▼
Defective late 60S maturation → impaired 40S–60S subunit joining
                       │
                       ▼
Fewer actively translating 80S ribosomes → ↓ bulk protein synthesis
                       │
          ┌────────────┴─────────────┐
          ▼                          ▼
↑ neuronal apoptosis          reduced neuronal output/proliferation
          │                          │
          └────────────┬─────────────┘
                       ▼
      Reduced brain growth  →  MICROCEPHALY (progressive)
      Disrupted neurodevelopment → ID, ASD, EPILEPSY, cerebellar signs
                       │
                       ▼ (tissue-specific translational vulnerability — branch)
Retinal degeneration (p.R32L) · Hearing loss + cardiac defects (p.K78E)
Genitourinary / GI / laryngeal anomalies · Growth failure

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CONTRAST — same gene, different disease:
SOMATIC RPL10 p.R98S (T-ALL, ~8%) → JAK-STAT gain-of-function,
↑ serine/glycine biosynthesis (PSPH), oncogenic mutagenesis (NOTCH1)
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The unifying principle is **dosage-sensitive translational insufficiency in a highly constrained ribosomal protein**: because *RPL10* is LoF-intolerant (pLI 0.997), only partial (hypomorphic) missense alleles are compatible with survival, and their effect manifests most severely in the high-translation-demand developing brain. The germline (hypomorphic, developmental) and somatic (R98S, oncogenic gain-of-function) diseases are cleanly dissociable — a valuable illustration of allele- and context-specific pathobiology.

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
25316788	*Novel ribosomopathy caused by dysfunction of RPL10… X-linked microcephaly*	Defines gene, XLR inheritance, LoF mechanism (zebrafish)	Human + model organism
16940977	*Mutations in RPL10 suggest a novel disease mechanism for autism*	Original RPL10 identification (L206M, H213Q)	Human clinical
9443083	*Assembly of the QM protein onto the 60S subunit… cytoplasm*	Normal uL16 function (subunit joining)	In vitro / yeast
35876338	*Postnatal microcephaly and retinal involvement expand phenotype*	Phenotype spectrum, ~50% microcephaly, conservation, gnomAD-absent	Human clinical
25846674	*RPL10 mutation segregating in family with XLID*	Per-feature frequencies in affected family	Human clinical
29066376	*De novo RPL10 mutation… syndromic ID and epilepsy*	Severe end (p.K78E) phenotype; review	Human clinical
40861044	*Novel hemizygous missense RPL10 (p.R116Q)*	New allele, maternal inheritance, in vitro LoF, expanded neonatal features	Human + in vitro
28744013	*RPL10 R98S enhances JAK-STAT signaling*	Distinct somatic oncogenic mechanism	Model / in vitro
30482776	*Ribosomal lesions promote oncogenic mutagenesis*	R98S recurrent in T-ALL; secondary mutagenesis	Model / in vitro
31186416	*Altered serine and glycine metabolism in T-ALL*	R98S metabolic rewiring (PSPH)	Model / in vitro
31090057	*MidXq28-duplication syndrome*	Multigenic Xq28 CNV context (FLNA, RPL10, GDI1)	Human clinical
20004760	*Dosage-dependent Xq28 copy-number gain*	GDI1 dosage as likelier CNV driver; skewed XCI	Human clinical

Reference databases used: OMIM (#300998, gene-MIM 312173), gnomAD v2 (constraint), UniProt P27635, PDB (≥37 human 80S ribosome structures), ClinVar (~404 records, NM_006013.5), NCBI Gene / Ensembl / HGNC, and MGI/ZFIN (orthologs).

Limitations and Knowledge Gaps

1. **Small N.** Fewer than ~20 molecularly confirmed males limit precise frequency estimates, genotype–phenotype correlation, and epidemiology (no reliable prevalence/incidence).
2. **VUS-dominant variant landscape.** Most *RPL10* missense variants in ClinVar are VUS; functional assays exist for only a few alleles (p.K78E, p.R116Q), so many candidate variants cannot be confidently classified.
3. **Mechanistic gaps.** The steps from reduced translation to specific clinical features (epilepsy, ASD, retinopathy, cardiac/GU anomalies) are largely inferred, not directly demonstrated in human neurons or organoids; the basis for tissue- and allele-specific involvement is unresolved.
4. **No mammalian disease model.** Published evidence rests on zebrafish morphants and yeast; a humanized knock-in mouse recapitulating the full syndrome is absent.
5. **No natural-history or quality-of-life data**, no biomarkers, and no therapeutic development specific to MRXS35.
6. **CNV vs point-mutation ambiguity.** Xq28 CNVs spanning *RPL10* involve multiple candidate genes (GDI1, FLNA), complicating attribution to *RPL10* itself.

Proposed Follow-up Experiments / Actions

1. **Establish a humanized knock-in mouse or human iPSC-derived neuron/organoid model** carrying p.K78E and p.R116Q to directly test the translation→apoptosis→microcephaly chain and dissect cell-type-specific vulnerability (neurons, photoreceptors, cardiomyocytes).
2. **Systematic functional assays** (subunit-joining, polysome profiling, ribosome footprinting) across all reported and VUS alleles to reclassify VUS and build a genotype–severity map — strengthening ACMG PS3 evidence.

3. **International patient registry / natural-history study** to quantify per-phenotype frequencies, onset, progression (especially microcephaly and retinopathy), survival, and quality of life.
4. **Translatome/single-cell profiling** of patient-derived neurons to identify the specific mRNAs whose translation is most sensitive to uL16 loss — candidate downstream effectors and biomarkers.
5. **Structural analysis** (cryo-EM/AlphaFold) mapping Arg32, Lys78, Arg116, Leu206, His213 onto the human 80S ribosome to model how each substitution perturbs rRNA binding and subunit joining.
6. **Explore translation-modulating or apoptosis-limiting therapeutic strategies** in the zebrafish/organoid models as proof-of-concept, given the absence of any disease-modifying therapy.

Report compiled from 5 completed investigation iterations, 7 confirmed findings, and 19 reviewed papers. Evidence types are distinguished as human clinical, model organism, in vitro, and computational throughout.
